MIL-R-19/1B
23 June 1967
SUPERSEDING
MIL-R-19/1A
16 December 1957

MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND

(LOW OPERATING TEMPERATURE), STYLE RA10

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RA10, low operating temperature, wire-wound, variable resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-19 - Resistors, Variable, Wire-wound (Low Operating Temperature), General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

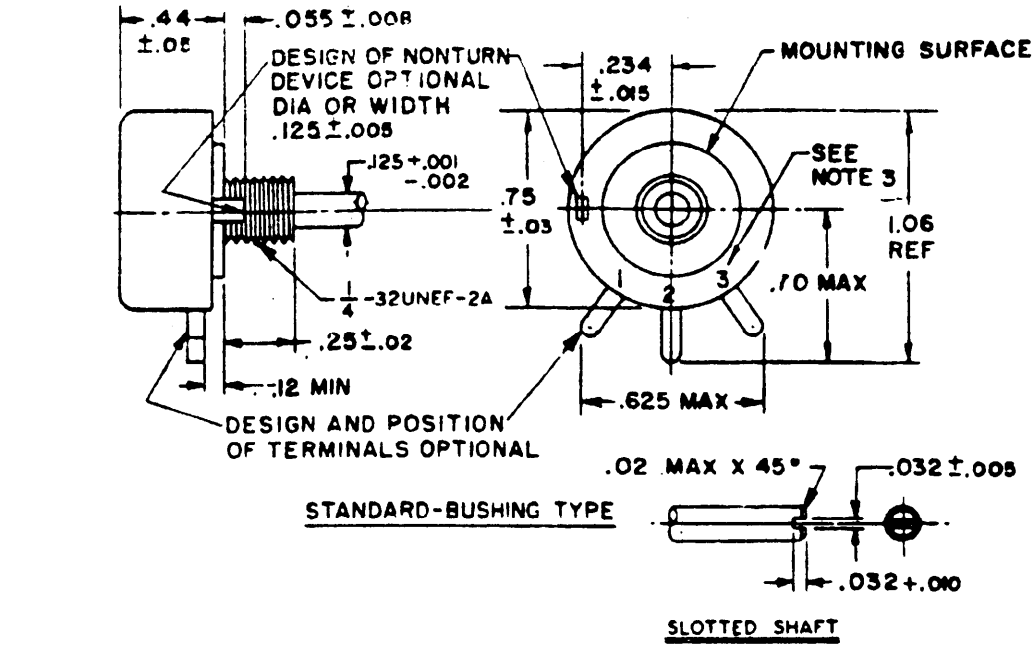
3. REQUIREMENTS

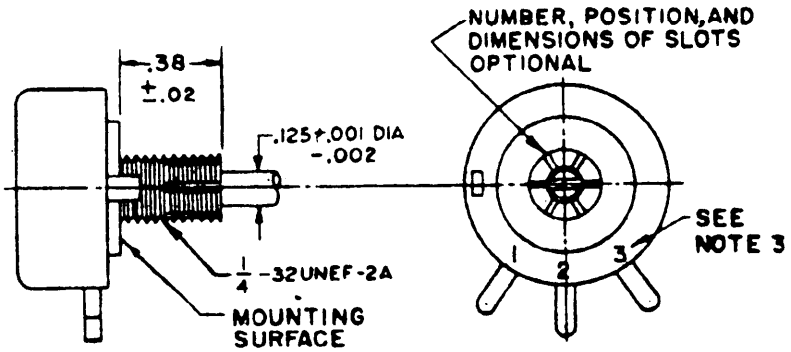
3.1 Requirements. Requirements shall be in accordance with MIL-R-19, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Resistance values and tapers. The resistance values and available tapers applicable to style RA10 resistors shall be as specified in table I.

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INCHES	
.001	
.002	.05
.005	.13
.008	.20
.010	.25
.015	.38
.02	.51
.03	.76
.032	.81
.05	1.27
.055	1.40
.12	3.05
.125	3.18
.234	5.94
.25	6.35
.38	9.65
.44	11.18
.625	15.88
.70	17.78
.75	19.05
1.06	26.92

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 2
3. Terminal identification for reference only.

FIGURE 1. Style RA10 resistors.

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TABLE I. Resistance values and tapers.

Nominal total resistance	Available tapers
Ohms	
15	A
25	A
50	A
75	A
100	A
150	A
200	A
250	A
350	A
500	A
750	A
1,000	A
1,500	A
2,000	A
2,500	A

3.4 Shaft length. The shaft length shall be as specified in table II.

TABLE II. Length of operating shaft.

Symbol	Length of shaft from mounting surface, nominal	
	Bushing N	Bushing L
	inches (± 1/32)	inches (± 1/32)
L	5/16	---
M	---	7/16
D	7/8	7/8

3.5 Power rating. The power rating shall be 1.0 watt at an ambient temperature of 40°C.

3.6 Hardware. The mounting nut shall be .062-inch thick and shall measure .312 inch across the hexagonal flats. The locking nut shall be .125-inch thick and shall measure .312 inch across the hexagonal flats. Thread size shall be 1/4-32UNEF-2B. Internal-tooth lockwasher of suitable size shall be supplied.

3.7 Torque. The torque required to effect rotation shall be not less than 0.3 ounce-inch nor more than 6 ounce-inches. The torque applied through the operating shaft to the stops shall be 3 pound-inches. For the locking-bushing type, the control arm shall not move when a torque of 20 ounce-inches is applied to the shaft after the locking nut is tightened with a torque of 8 pound-inches.

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3.8 Total mechanical and electrical rotation. The total mechanical and electrical rotation shall be 290° to 305°. Switches are not applicable to this specification.

3.9 Retainer rings. If retainer rings are used, they shall be not thicker than .025 inch.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-19.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-19.

6. NOTES

6.1 Notes. The notes specified in MIL-R-19 are applicable to this specification.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MI, MU
Navy - AS, OS, SH
Air Force - 11, 17, 85

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 14, 19

Preparing activity:
Army - EL

(Project 5905-0505-2)

Code "C"

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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